

Jacopo Bilotto

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Computational physicist and engineer specializing in differentiable physics simulation, scientific ML and HPC. Developed GPU-accelerated JAX solvers and hybrid physics-neural models for inverse design, alongside high-performance C++ simulation capabilities and large-scale HPC workflows. Industry experience at Bühler and ASML translating models into efficient engineering tools. Experienced in Python, C++, and scientific computing.

Experience, Roles and Responsibilities

EPFL, in partnership with Bühler AG

Lausanne, Switzerland

PHD RESEARCHER IN COMPUTATIONAL SCIENCE

Mar 2023 – Present

- Developed a hybrid MLP-differentiable mechanics framework for **inverse design** of tribological metainterfaces; generated a 100k-sample synthetic dataset, designed physics-based losses and curriculum training, and validated predictions against high-fidelity BEM simulations. Predicted new designs with more than **30x speedup** compared to state of the art.
- Engineered a high-performance **differentiable GPU EHL solver** in **JAX**, allowing high resolution analysis (~300M DOFs) and optimization. Reduced computational time from hours to seconds.
- Automated a large-scale DEM simulation campaign on a Linux HPC cluster, using Latin Hypercube Sampling (LHS) to train a **GPR surrogate model** and perform Pareto analysis, significantly reducing computational overhead and operational costs (improved process efficiency by 2-10%).
- Engineered and merged new physics support into the core LAMMPS C++ repository in collaboration with Sandia Labs researchers, expanding simulation capabilities of the **largest molecular dynamics code** for thousands of global users.
- Technical lead & mentor of 9 student research projects, determining project timelines and methodology for topics ranging from Reduced Order Models to **Graph Neural Simulators**.
- Teaching assistant for multiple university courses covering numerical methods and computational physics.
- Coordinated international technical collaborations spanning industry partners, Sandia National Labs (USA), Lund University (Sweden), and the University of Tokyo (Japan) to align objectives and mitigate project risks.

Mechanics Gathering Seminar (MEGA)

Lausanne, Switzerland

LEAD ORGANIZER & MEMBER

Feb 2024 - Present

- Lead organizer for two semesters, coordinating weekly technical seminars for early-career researchers in mechanics at EPFL. Increased participation from 10 to 30 people.

LauzHack (EPFL Hackathon)

Lausanne, Switzerland

BOARD MEMBER & SPONSORING LEAD

Apr 2023 - Nov 2025

- Managed corporate partnerships and contributed to raising **60K CHF annually** from sponsors including AWS, Logitech, and Bristol Myers Squibb.

ASML

Eindhoven, Netherlands

SIMULATION INTERN

Apr 2022 – Dec 2022

- Validated computational models against experimental data, achieving an output match with **<5% error**, and deployed automated code to the design team for continuous performance assessments.
- Engineered a computationally efficient analytical model to enable rapid design assessment of complex architectures, replacing heavy numerical simulations and costly physical experiments.
- Identified and resolved a critical numerical instability within the core solver formulation in collaboration with the software development team.

INFN (National Institute of Nuclear Physics)

RESEARCH & TECHNICAL STUDENT

Frascati, Italy

Sep 2021 – Nov 2021

- Developed LabVIEW control systems for electrical cannons, optimizing hardware performance and reducing filament thermal stress.
- Reduced experimental data post-processing time by **70%** through the development of automated Python-based workflows, streamlining operational efficiency.

Technical Skills

Programming Python (NumPy, JAX, PyTorch), C++ (MPI), MATLAB
Tools Git, Linux/Unix (Bash), CMake, GitHub (CI/CD), Unit Testing, OpenCode, ParaView
Simulation Abaqus (User Subroutines), Ansys, COMSOL, LAMMPS (MD/DEM), Tamaas (BEM)
Languages Italian (Native), English (C1/C2), French (B2)

Education

EPFL

PHD IN MECHANICS

Switzerland

Expected Feb 2027

Recipient of the **EPFL GlobalLeaders Fellowship** (MSCA programme, €105,000).

Sapienza, University of Rome

M.S. & B.S. IN MECHANICAL ENGINEERING, COMPUTATIONAL SCIENCE TRACK

Italy

Sep 2017 - Jan 2023

Cum Laude (GPA: 29.8/30). Ranked in the top 3 students of the year.

University College Lamaro Pozzani

GLOBAL STUDIES & FUNDAMENTALS OF BUSINESS

Rome, Italy

Sep 2017 - Jul 2022

Fully funded scholarship equivalent to €60,000. Interdisciplinary honors program covering: Business Fundamentals, Political Systems, Game Theory, Demography, and Economic Theory.

Selected Publications

- **J. Bilotto**, J. Kolinski, B. Lecampion, J.-F. Molinari, G. Subhash, J. Garcia-Suarez, “Fluid-mediated impact of soft solids,” *Journal of Fluid Mechanics* (2024), vol. 997, A35, doi:10.1017/jfm.2024.820
- **J. Bilotto**, M. Trulsson, J.-F. Molinari, “Shear flow of frictional spheroids: Comparison between elongated and flattened particles,” *Physical Review E* (2025), vol. 112, doi:10.1103/tj41-6qqk
- (Accepted at *Tribology Letters*) **J. Bilotto**, A. Singhal, G. Cortes, J. Garcia-Suarez, L. Fourel, J.-F. Molinari, “Inverse Design of Metainterfaces for Static Friction Control: Beyond the Hertzian Limit”, arxiv 2026, doi: 10.48550/arXiv.2605.11012
- (In preparation) **J. Bilotto**, J.-F. Molinari, “A fast GPU differentiable solver for elastohydrodynamic lubrication”